Matrix First Generation

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THE GOAL IS TO GENERATE A NICE 2x2 MATRIX FIRST AND USE THAT TO DETERMINE WHERE THE SHIP POSITION SHOULD BE AND ASSOCIATED MULTIPLYING VECTOR

- LIMITATIONS:
  MATRIX VECTORS SHOULDN'T BE TOO LARGE
  VECTORS CANNOT BE PARALLEL OR TOO CLOSE PARALLEL

COMPONENTS:

$pos: \begin{bmatrix} x \\ y \end{bmatrix}$ $VECT: \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ $MATRIX: \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (X_1, X_2)

FORMULA:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} = \alpha\begin{bmatrix} a \\ c \end{bmatrix} + \beta\begin{bmatrix} b \\ d \end{bmatrix}$$

(X_1, X_2)

GENERATE X_1 BY SOME RANDOM CONSTRAINTS: $R_{MAX} = 5$, $R_{MIN} = -5$

CHOOSE $a = 3$ $c = 2$

EDGE CASE NOTE: IF $a = c = 0$, RESTART
WE CAN SWAP X_1 AND X_2 LATER IF WANTED BUT WE NEED NON-ZEROS TO START

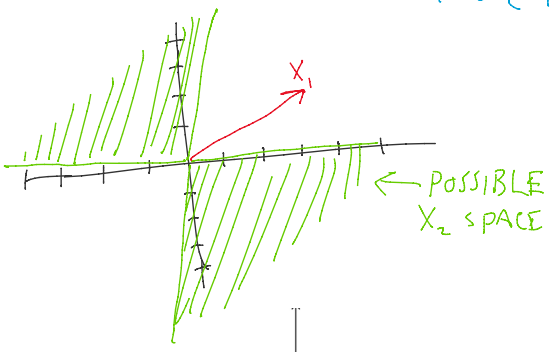
$$\alpha\begin{bmatrix} 3 \\ 2 \end{bmatrix} + \beta\begin{bmatrix} c \\ d \end{bmatrix}$$

CHOOSE A TEMPORARY VECTOR $k^{(k)}$ AS SOME 2D ROTATION OF X_1 NORMALIZED AROUND A RANDOMLY GENERATED θ WITH CONSTRAINTS $\theta_{MAX} = 150$, $\theta_{MIN} = 40$ AND A POSITIVE/NEGATIVE COINFLIP: $\pm$

NOTE: WE MAY ALSO DEFINE θ AS SOME SET OF MULTIPLES FOR SOME RANGE
ie $\theta = \{40, 50, 60, \ldots 130, 140\}$

$$k = \begin{bmatrix} s \\ t \end{bmatrix} \qquad X_1 \cdot UNIT \begin{bmatrix} a_u \\ c_u \end{bmatrix}$$

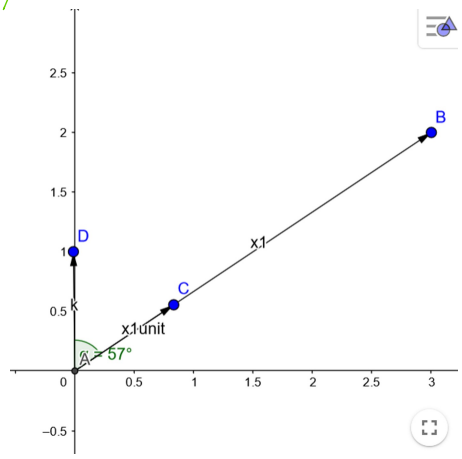
$$\begin{bmatrix} s \\ t \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} a_u \\ c_u \end{bmatrix}$$



CHOOSE $\pm = +$ and $\theta = 57$

$$\begin{bmatrix} \cos(57) & -\sin(57) \\ \sin(57) & \cos(57) \end{bmatrix} \qquad X_1 \cdot UNIT = \begin{bmatrix} 3/\sqrt{13} \\ 2/\sqrt{13} \end{bmatrix}$$

$$k = \begin{bmatrix} s \\ t \end{bmatrix} = \begin{bmatrix} 3/\sqrt{13} \cdot \cos(57) - 2/\sqrt{13}\sin(57) \\ 3/\sqrt{13} \cdot \sin(57) + 2/\sqrt{13} \cdot \cos(57) \end{bmatrix} = \begin{bmatrix} -0.01204 \\ 0.99993 \end{bmatrix}$$



NOW WE NEED TO FIND THE BEST SCALAR FOR k THAT GIVES US AS CLOSE AS POSSIBLE TO INTEGERS BUT IS STILL WITHIN OUR R VALUES

TO DO THIS WE'LL TEST EACH VALUE FROM 1 TO R_{MAX} AND SCORE IT. THE SCORE WILL BE THE AVG BETWEEN THE DIFFERENCES OF $|ROUND(R \cdot K) - (R \cdot K)|$ mxn : 2x1

$\hat{s}: + \hat{t}$, THINKING LARGER

OF $|\text{ROUND}(R \cdot K) - (R \cdot K)|$

$$\widehat{\text{SCORE}_i} = \left| \text{ROUND}\left(R_i \begin{bmatrix} s \\ t \end{bmatrix}\right) - R_i \begin{bmatrix} s \\ t \end{bmatrix} \right| \rightarrow \text{SCORE}_i = \frac{\hat{s}_i + \hat{t}_i}{2}$$

THINKING LARGER

$$\text{SCORE}_i = \frac{\sum_1^m \hat{s}_i}{m}$$

PICK R_i WHERE SCORE_i IS MINIMUM

HERE WHEN $R_i = 1$ SCORE IS SMALLEST SO CHOOSE $R = 1$ THEN FINALLY

$$X_2 = \text{ROUND}(R \cdot k)$$

R_i	SCORE_i
1	0.006055
2	0.01211
3	0.018165
4	0.02422
5	0.030275

NOTE THIS SCORING MAY RESULT IN SOME BIASED DISTRIBUTION OF R. IN THAT CASE A WEIGHTED PROBABILITY MAY BE USED TO HELP KEEP THE CHOICE OF R MORE VARIED

SO OUR MATRIX IS

$$\begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \alpha \begin{bmatrix} 3 \\ 2 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

NOW WE HAVE TO PICK A VECT $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ THAT RESULTS IN A POS $\begin{bmatrix} x \\ y \end{bmatrix}$ THAT STATISFIES THESE REQUIREMENTS:

$$-75 \le x \le 75 \text{ and } 15 \le y \le 100$$

OPTIONS:

- RANDOMLY PICK $\begin{bmatrix} x \\ y \end{bmatrix}$ FIRST AND SOLVE FOR $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
- PICK $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ and test IF $\begin{bmatrix} x \\ y \end{bmatrix}$ IS VALID

PICKING $\begin{bmatrix} x \\ y \end{bmatrix}$

ITS EASIER CONCEPTUALLY TO PICK $\begin{bmatrix} x \\ y \end{bmatrix}$ FIRST BECAUSE WE KNOW THE CONSTRAINTS AHEAD OF TIME. HOWEVER GETTING A COMPUTER TO SOLVE A SYSTEM MAY BE MORE CHALLENGING

GENERATE A RANDOM SHIP POS $\begin{bmatrix} x \\ y \end{bmatrix}$ IN THE PLAY FIELD

HOPEFULLY THERE'S SOME LIBRARY FUNCTION FOR THIS

ex) CHOOSE $\begin{bmatrix} -60 \\ 83 \end{bmatrix}$

$$\begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} -60 \\ 83 \end{bmatrix}$$

$$\alpha = -20$$

$$\beta = 123$$

$$\text{VECT} = \begin{bmatrix} -20 \\ 123 \end{bmatrix}$$

$\rightarrow$

$$\begin{array}{cc} \alpha & \beta \end{array}$$

$$\left[\begin{array}{cc|c} 3 & 0 & -60 \\ 2 & 1 & 83 \end{array}\right]$$

COMP CAN'T JUST "CHOOSE 1/3"

$R_1 = R_1 \cdot 1/3$

$$\left[\begin{array}{cc|c} 1 & 0 & -20 \\ 2 & 1 & 83 \end{array}\right]$$

$R_2 = R_1 \cdot (-2) + R_2$

$$\left[\begin{array}{cc|c} 1 & 0 & -20 \\ & 1 & 123 \end{array}\right]$$

VECT = $\begin{bmatrix} -20 \\ 123 \end{bmatrix}$

OUR SOLUTION THEN IS

$\begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} -20 \\ 123 \end{bmatrix} = \begin{bmatrix} -20 \\ 123 \end{bmatrix}$

$\left[\begin{array}{cc|c} 1 & 0 & -20 \\ 0 & 1 & 123 \end{array}\right]$

EDGE CASE: THE SYSTEM HAS NO SOLUTIONS IF $\alpha = \beta = 0$ IN THIS CASE THROW AWAY $\begin{bmatrix} x \\ y \end{bmatrix}$ AND RESTART

(2x2)

CHOOSE $\begin{bmatrix} 37 \\ 79 \end{bmatrix}$

$\begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 37 \\ 79 \end{bmatrix}$

$\begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 37/3 \\ 163/9 \end{bmatrix} = \begin{bmatrix} 37 \\ 79 \end{bmatrix}$

↑ BUT THIS FRACTIONAL WONT WORK SO WE ROUND IT AND READJUST $\begin{bmatrix} x \\ y \end{bmatrix}$

$\begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 12 \\ 54 \end{bmatrix} = \begin{bmatrix} 36 \\ 78 \end{bmatrix}$

$\left[\begin{array}{cc|c} 3 & 0 & 37 \\ 2 & 1 & 79 \end{array}\right]$

$R_1 = R_1 \cdot \frac{1}{3}$

$\left[\begin{array}{cc|c} 1 & 0 & 37/3 \\ 2 & 1 & 79 \end{array}\right]$

$R_2 = R_1 \cdot (-2) + R_2$

$\left[\begin{array}{cc|c} 1 & 0 & 37/3 \\ 0 & 1 & 163/9 \end{array}\right]$

SOME THINGS OF NOTE:

TO GET SMALLER $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ VALUES WE NEED TO INCREASE R

HOWEVER THE LARGER R IS THE LARGER THE LATTICE MATRIX WILL BE.

SIMILARLY, AS R INCREASES THE PROBLEM BECOMES MORE COMPLEX

- CONSIDER R = 20 THEN $\begin{bmatrix} 18 & -15 \\ 6 & 12 \end{bmatrix}$ IS A VALID MATRIX

$\begin{bmatrix} 18 & -15 \\ 6 & 12 \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} -64 \\ 53 \end{bmatrix}$

$\left[\begin{array}{cc|c} 18 & -15 & -64 \\ 6 & 12 & 53 \end{array}\right]$

$R_1 = R_2 \cdot (-3) + R_1$

$\left[\begin{array}{cc|c} 0 & -51 & -223 \\ 6 & 12 & 53 \end{array}\right]$

$R_1 = R_1 \cdot (-\frac{1}{51})$

$R_2 = R_2 \cdot (\frac{1}{6})$

$\left[\begin{array}{cc|c} 0 & 1 & 4.372 \\ 1 & 2 & 8.833 \end{array}\right]$

$R_2 = R_1 \cdot (-2) + R_2$

$R_1 \rightleftarrows R_2$

$\left[\begin{array}{cc|c} 1 & 0 & 0.088 \\ 0 & 1 & 4.372 \end{array}\right]$

AFTER ROUNDING $\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$

READJUSTING POS WE GET

$\begin{bmatrix} 18 & -15 \\ 6 & 12 \end{bmatrix}\begin{bmatrix} 0 \\ 4 \end{bmatrix} = \begin{bmatrix} -60 \\ 48 \end{bmatrix}$

SO NOT ONLY DO HIGHER R VALUES RESULT IN MORE COMPLEX MATRIX MATH FOR THE PLAYER TO DO IN THEIR HEAD, BUT THE ROUNDING ERROR BECOME MUCH

SO NOT ONLY DO HIGHER R VALUES RESULT IN MORE [illegible] MATH FOR THE PLAYER TO DO IN THEIR HEAD, BUT THE ROUNDING ERROR BECOME MUCH LARGER AS WELL

PICK $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ SINCE WE DON'T KNOW AHEAD OF TIME WHAT $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ WILL BE WE DONT HAVE ANY HARD RESTRAINTS FOR $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

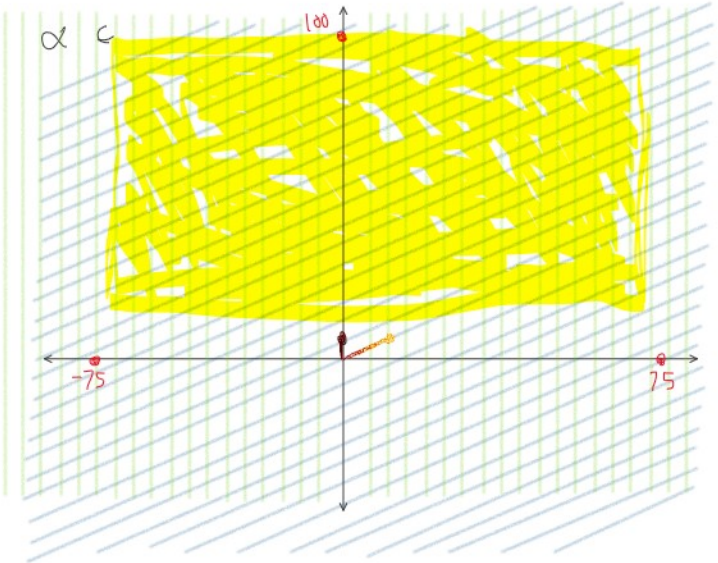
BUT AFTER $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ IS GENERATED WE CAN USE IT AND THE RESTRAINTS FOR $\begin{bmatrix} x \\ y \end{bmatrix}$ TO DYNAMICALLY GENERATE RESTRAINTS FOR $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

THIS SEEMS LIKE ITS MORE WORK OR HARDER BUT COMPUTATIONALLY IT'S FASTER AND DOESN'T RESULT IN ROUNDING ERRORS

MATRIX $= \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}$ $\quad -75 \le x \le 75$ $\quad 15 \le y \le 100$

$$x = \alpha a + \beta b$$
$$y = \alpha c + \beta d$$

So $\quad \alpha \begin{bmatrix} 3 \\ 2 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$



- ORIGINAL CARTESIAN PLANE
- X_1 SPANS
- X_2 SPANS
- X_1
- X_2
- POSSIBLE POINTS FOR $\begin{bmatrix} x \\ y \end{bmatrix}$

SO WHAT DO I KNOW!

$$X_{min} \le \alpha a + \beta b \le X_{MAX} \quad = \quad -75 \le \alpha 3 + \beta 0 \le 75$$
$$y_{min} \le \alpha c + \beta d \le y_{max} \quad \quad 15 \le \alpha 2 + \beta 1 \le 100$$

I HAVE A RECTANGLE WITH SIDES OF $(\alpha 3 + \beta 0) \times (\alpha 2 + \beta 1)$

WHAT IS THE MAX VALUE WE CAN MULTIPLY X_1 BY AND BE IN THE RECTANGLE?

WHAT IS THE MIN VALUE?

WHAT ABOUT FOR X_2?

LETS DO SOME BOUNDARY CHECKING:

FOR X_1 FOR $a \in x$ $\quad \alpha_x \ge \frac{x_{min}}{a} = \frac{-75}{a} = \frac{-75}{3} \Rightarrow \alpha_x \ge -25$

and $\alpha_x \le \frac{x_{max}}{a} = \frac{75}{a} = \frac{75}{3} \Rightarrow \alpha_x \le 25$

FOR $c \in y$ $\quad \alpha_y \ge \frac{y_{min}}{c} = \frac{15}{c} = \frac{15}{2} \Rightarrow \alpha_y \ge 7.5 \approx \alpha_y \ge 8$ ← CEIL ONLY

$\alpha_y \le \frac{y_{MAX}}{c} = \frac{100}{c} = \frac{100}{2} \Rightarrow \alpha_y \le 50$

FOR X_1 $\quad -25 \le \alpha_x \le 25$

$\quad 8 \le \alpha_y \le 50$

$> \frac{-75}{3}$ ← UH OH

FOR THIS CASE ITS IRRELEVANT WHAT THE BOUNDS

FOR X_1 $-25 \le \alpha_x \le 25$

$8 \le \alpha_y \le 50$

FOR X_2 FOR $b \in X$ $\beta_x \ge \frac{-75}{0}$ ← UH OH FOR THIS CASE ITS IRRELEVANT WHAT THE BOUNDS OF β_x ARE BECAUSE P_x IS NULL

$\beta_x \le \frac{75}{0}$

$d \in y$ $\beta_y \ge \frac{15}{d} = \frac{15}{1} \Rightarrow \beta_y \ge 15$

$\beta_y \le \frac{100}{d} = \frac{100}{1} \Rightarrow \beta_y \le 100$

FOR X_2 β_x IS NULL

$15 \le \beta_y \le 100$

SO NOW I KNOW THAT FOR α ITS RANGE OF POSSIBLE VALUES ARE $\underbrace{8 \le \alpha \le 25}_{\text{BOTH X Y NARROW SET}}$ OR $\underbrace{-25 \le \alpha \le 50}_{\text{X or Y BROAD SET}}$

AND FOR β $\underbrace{15 \le \beta \le 100}_{\text{NARROW SET}}$ $\underbrace{-\infty \le \beta \le \infty}_{\text{BROAD SET}}$ ← NOTE FOR PRACTICAL PURPOSES WHEN THIS HAPPENS WE WILL ASSIGN BOUNDS AS SOME PREDETERMINED MAX VALUE.

BUT NOW WE HAVE A SET OF RANGES TO WORK FROM

START BY PICKING FROM α OR β VIA COINFLIP

CHOOSE α

COINFLIP TO PICK THE BROAD OR NARROW SET

CHOOSE BROAD

α = RANDOM RANGE $(-25, 50)$

CHOOSE $\alpha = -11$

$$(-11)\begin{bmatrix}3\\2\end{bmatrix} + \beta\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}x\\y\end{bmatrix}$$

$$\begin{bmatrix}-33\\-22\end{bmatrix} + \beta\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}x\\y\end{bmatrix}$$

OR MAYBE JUST ONLY DEFINE RANGES FOR THE SECOND HALF AFTER THE FIRST IS CHOSEN?

GIVEN $\alpha = -11$ WE NEED TO REDEFINE OUR RANGES FOR β

FOR X_2 FOR $b \in X$ $\beta_x \ge \frac{x - \alpha a}{c} = \frac{-75 - (-33)}{0}$ ← AGAIN β_x IS ~~UNBOUND~~ NULL OR AND ARBITRARY

$\beta_x \le \frac{x - \alpha a}{c} = \frac{75 - (-33)}{0}$

$d \in y$ $\beta_y \ge \frac{y - \alpha c}{d} = \frac{15 - (-22)}{1} \Rightarrow \beta_y \ge 37$

$\beta_y \le \frac{y - \alpha c}{d} = \frac{100 - (-22)}{1} \Rightarrow \beta_y \le 122$

SO THE NEW NARROW SET FOR β IS $37 \le \beta \le 122$

GIVEN $\alpha = -11$ and $\begin{bmatrix}3 & 0\\2 & 1\end{bmatrix}$ any NUMBER FROM THIS SET SHOULD EXIST IN OUR RECTANGLE

WE'LL TRY BOTH TO TEST

CHOOSE $\beta = 37$

$$\begin{bmatrix}3 & 0\\2 & 1\end{bmatrix}\begin{bmatrix}-11\\37\end{bmatrix} = \begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}-33\\15\end{bmatrix}$$ ✓

CHOOSE $\beta = 122$

$$\begin{bmatrix}3 & 0\\2 & 1\end{bmatrix}\begin{bmatrix}-11\\122\end{bmatrix} = \begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}-33\\100\end{bmatrix}$$ ✓

$$\begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} -11 \\ 122 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cdots \\ 100 \end{bmatrix}$$ ✓

EDGE CASE:

WHAT IF WE PICK A VALUE FOR α THAT IS VALID IN ITS BROAD SET BUT IMPOSSIBLE TO FIND A SOLUTION FOR β

CHOOSE $\alpha = 35$

$$35\begin{bmatrix} 3 \\ 2 \end{bmatrix} + \beta\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 105 \\ 70 \end{bmatrix} + \beta\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\frac{-75-105}{0} \le \beta_x \le \frac{75-105}{0} \qquad \frac{15-70}{1} \le \beta_y \le \frac{100-70}{1}$$

$$-\infty \le \beta_x \le \infty \qquad -55 \le \beta_y \le 30$$

BROAD — NARROW

AT THIS POINT THERE IS NO VALUE FOR β WHERE X IS WITHIN OUR RANGE

SOLUTIONS:

WHEN DETERMINING INTIAL BOUNDS FOR α and β FLAG WHEN A SET IS NULL

IN THIS CASE β_x IS NULL SO THE BROAD SET OF VALUES FOR α CANNOT HAVE VALUES OUTSIDE OF α_x

SO WE REDEFINE OUR BROAD SET AS α_a

IF β_y IS NULL THEN α MUST BE DEFINED BY α_y

RECAP AND FORMULAS

$$\frac{x_{min}}{a} \le \alpha_x \le \frac{x_{max}}{a} \qquad \frac{y_{min}}{c} \le \alpha_y \le \frac{y_{max}}{c}$$

$$\frac{x_{min} - \alpha a}{b} \le \beta_x \le \frac{x_{max} - \alpha a}{b} \qquad \frac{y_{min} - \alpha c}{d} \le \beta_y \le \frac{y_{max} - \alpha c}{d}$$